Sampling of Continuous-Time Signals

CSE 219: Signals and Linear Systems

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1 Introduction

Sampling is the fundamental process of converting a continuous-time signal into a discrete-time sequence. A physical continuous-time signal $x(t)$ contains an infinite, uncountable number of time points across any finite interval. Because digital hardware (such as computers and DSP processors) possesses finite memory and processing bandwidth, storing or computing infinitely dense continuous signals is impossible. Consequently, we must discretize the signal in time by acquiring distinct samples at discrete intervals, which can then be processed digitally.

A primary challenge in sampling is the potential loss of information. When converting $x(t)$ into a sampled sequence, we discard the continuous continuum of values between samples. This naturally raises a critical question: *Can we perfectly reconstruct the original continuous-time signal $x(t)$ from its discrete samples?*

If the continuous signal changes slowly relative to the interval between consecutive samples, adjacent samples will retain a high degree of correlation, enabling perfect signal recovery. Conversely, if the samples are spaced too far apart, rapid variations in $x(t)$ between sampling instances are lost, forcing us to guess the intermediate behavior—which often leads to severe distortion. Intuitively, a higher sampling rate decreases the time gap between consecutive samples, preserving the fine structure of the signal and allowing accurate reconstruction.

However, increasing the sampling rate indefinitely imposes unnecessary computational load. What is the minimum acceptable rate required to reconstruct $x(t)$ without loss of information? The celebrated **Nyquist-Shannon Sampling Theorem** provides the definitive answer: to guarantee exact signal recovery, a band-limited signal must be sampled at a rate strictly greater than twice its highest frequency component (known as the *Nyquist rate*).

2 Mathematical Foundations & Mathematical Tools

Before analyzing sampling in detail, we review several essential continuous-time Fourier transform (CTFT) properties and signal operations.

2.1 The Duality Property

Let $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$ be a continuous-time Fourier transform pair. The inverse Fourier transform is defined as:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, d\omega \tag{1}$$

By interchanging the roles of time t and frequency ω, we apply a change of variables. Let $t \to -\omega$ and $\omega \to t$. Substituting these into Equation (1) yields:

$$x(-\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jt) e^{-j\omega t}\, dt \tag{2}$$

Multiplying both sides by 2π:

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(jt) e^{-j\omega t}\, dt \tag{3}$$

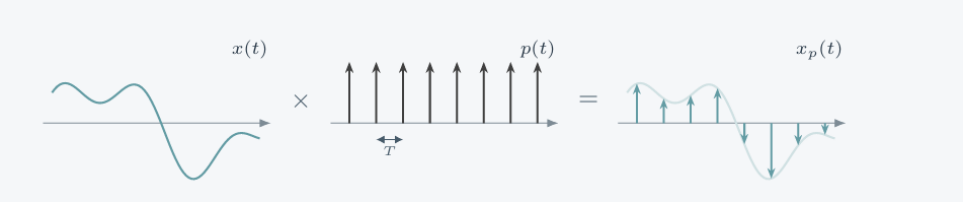
Notice that the right-hand side represents the forward Fourier transform of the time-domain signal $X(jt)$. Hence, we establish the **Duality Property**:

$$X(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x(-\omega) \tag{4}$$

2.2 Mechanism of Sampling: Impulse Multiplication

Recall the sampling property of the Dirac delta function $\delta(t)$: multiplying any continuous signal $x(t)$ by an instantaneous impulse located at $t = t_0$ extracts the signal's value at that specific instant:

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0) \tag{5}$$



To sample a signal periodically across all time, $x(t)$ is multiplied by an **impulse train** $p(t)$ with period T_s (where T_s is the sampling period):

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \tag{6}$$

The resulting sampled signal $x_p(t)$ is expressed as:

$$\begin{aligned} x_p(t) &= x(t) \cdot p(t) \\ &= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \\ &= \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s) \end{aligned} \tag{7}$$

2.3 Convolution and Multiplication Dualities

A fundamental principle of Fourier analysis is the dual relationship between time-domain and frequency-domain operations.

Convolution in Time Domain: Convolution in the time domain corresponds to point-wise multiplication in the frequency domain:

$$x_1(t) * x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} X_1(j\omega)X_2(j\omega) \tag{8}$$

Multiplication in Time Domain: Conversely, point-wise multiplication in the time domain corresponds to continuous convolution in the frequency domain scaled by $\frac{1}{2\pi}$.

Proof. Let $u(t) = X_1(jt)$ and $v(t) = X_2(jt)$. By the Duality Property:

$$\begin{aligned} u(t) = X_1(jt) &\overset{\mathcal{F}}{\longleftrightarrow} U(j\omega) = 2\pi x_1(-\omega) \\ v(t) = X_2(jt) &\overset{\mathcal{F}}{\longleftrightarrow} V(j\omega) = 2\pi x_2(-\omega) \end{aligned}$$

Using time-domain convolution on $x_1(t) * x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} X_1(j\omega)X_2(j\omega)$ and invoking duality:

$$u(t) \cdot v(t) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\Big(x_1(-\omega) * x_2(-\omega)\Big) \tag{9}$$

Expressed directly in terms of $U(j\omega)$ and $V(j\omega)$:

$$\begin{aligned} u(t) \cdot v(t) &\overset{\mathcal{F}}{\longleftrightarrow} 2\pi \cdot \left(\frac{1}{2\pi}U(j\omega)\right) * \left(\frac{1}{2\pi}V(j\omega)\right) \\ u(t) \cdot v(t) &\overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi}\Big[U(j\omega) * V(j\omega)\Big] \end{aligned} \tag{10}$$

□

2.4 Frequency Shifting Property of the CTFT

Let $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$ be a Fourier transform pair. Multiplying a time-domain signal by a complex exponential $e^{j\omega_0 t}$ results in a shift of its frequency spectrum by ω_0:

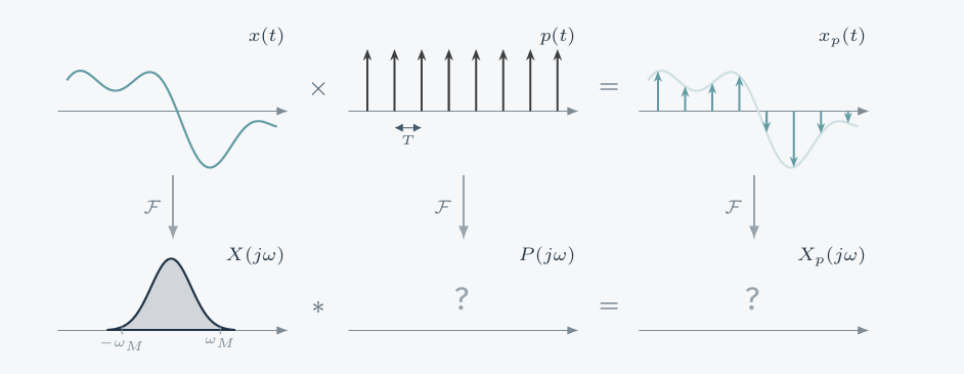
$$e^{j\omega_0 t}x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j(\omega - \omega_0)) \tag{11}$$

Proof. By the definition of the continuous-time Fourier transform:

$$\begin{aligned}\mathcal{F}\left\{e^{j\omega_0 t}x(t)\right\} &= \int_{-\infty}^{\infty} \left(e^{j\omega_0 t}x(t)\right) e^{-j\omega t}\, dt \\ &= \int_{-\infty}^{\infty} x(t)e^{-j(\omega-\omega_0)t}\, dt \\ &= X(j(\omega - \omega_0))\end{aligned} \tag{12}$$

□

3 Frequency Spectrum of the Sampled Signal



To understand how sampling alters a signal in the frequency domain, we examine the Continuous-Time Fourier Transform (CTFT) of the impulse-sampled signal $x_p(t)$. Taking the direct Discrete-Time Fourier Transform (DTFT) of discrete samples $x[n]$ alone does not immediately reveal the explicit structural relationship between the original spectrum $X(j\omega)$ and the sampled spectrum $X_p(j\omega)$.

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x(nT)e^{-j\omega nt} \tag{13}$$

Instead, we treat impulse sampling as continuous-time multiplication: $x_p(t) = x(t) \cdot p(t)$. By transforming both $x(t)$ and $p(t)$ into the frequency domain, we can evaluate their continuous convolution:

$$X_p(j\omega) = \frac{1}{2\pi}\Big[X(j\omega) * P(j\omega)\Big] \tag{14}$$

While the spectrum $X(j\omega)$ of a band-limited input signal is assumed known, deriving the Fourier Transform of the infinite impulse train $p(t)$ requires care.

3.1 Fourier Transform of the Periodic Impulse Train

$$P(j\omega) = \sum_{n=-\infty}^{\infty} e^{-j\omega nt} \tag{15}$$

Recall the definition of the periodic impulse train with period T_s:

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \tag{16}$$

Because $p(t)$ is a periodic function with fundamental period T_s and fundamental frequency $\omega_s = \frac{2\pi}{T_s}$, it can be expanded into a continuous-time Fourier series (CTFS):

$$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_s t} \tag{17}$$

The Fourier series coefficients a_k are evaluated over one fundamental period $\left[-\frac{T_s}{2}, \frac{T_s}{2}\right]$:

$$\begin{aligned} a_k &= \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} p(t) e^{-jk\omega_s t}\, dt \\ &= \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-jk\omega_s t}\, dt \end{aligned} \tag{18}$$

Applying the sifting property of the impulse function at $t = 0$:

$$a_k = \frac{1}{T_s} e^0 = \frac{1}{T_s} \quad \forall k \in \mathbb{Z} \tag{19}$$

Thus, the continuous-time impulse train can be written in exponential Fourier series form as:

$$p(t) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{jk\omega_s t} \tag{20}$$

Now, we transform $p(t)$ into the frequency domain. Recalling the CTFT transform pairs:

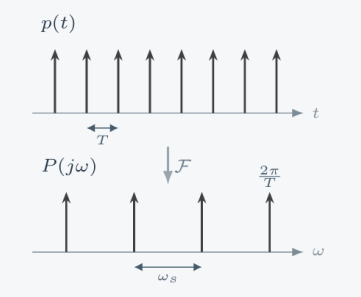
$$\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1 \tag{21}$$

$$1 \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega) \tag{22}$$

$$e^{jk\omega_s t} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega - k\omega_s) \tag{23}$$

Taking the Fourier transform of $p(t)$ term-by-term yields $P(j\omega)$:

$$P(j\omega) = \mathcal{F}\{p(t)\} = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \tag{24}$$

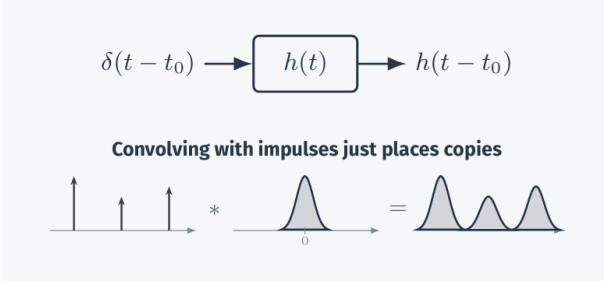


This demonstrates that the Fourier transform of a time-domain impulse train with period T_s is another impulse train in the frequency domain with period $\omega_s = \frac{2\pi}{T_s}$ and impulse weights of $\frac{2\pi}{T_s} = \omega_s$.

3.2 Derivation of the Sampled Spectrum $X_p(j\omega)$

With $P(j\omega)$ established, we compute the spectrum of the sampled signal $x_p(t) = x(t) \cdot p(t)$ using frequency-domain convolution:

$$\begin{aligned} X_p(j\omega) &= \frac{1}{2\pi}\Big[X(j\omega) * P(j\omega)\Big] \\ &= \frac{1}{2\pi}\left[X(j\omega) * \left(\frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)\right)\right] \\ &= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \Big[X(j\omega) * \delta(\omega - k\omega_s)\Big] \end{aligned} \tag{25}$$

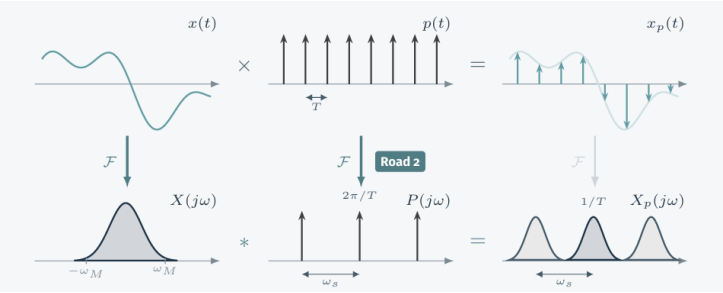


Convolving any continuous spectrum with a shifted impulse simply shifts the spectrum to the location of the impulse: $X(j\omega) * \delta(\omega - k\omega_s) = X(j(\omega - k\omega_s))$. Therefore, we arrive at the fundamental equation of impulse sampling:

Spectrum of an Impulse-Sampled Signal

$$X_p(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \tag{26}$$

Physical Interpretation: The frequency spectrum $X_p(j\omega)$ of the sampled signal consists of an infinite sum of periodically shifted, overlapping copies of the original signal spectrum $X(j\omega)$. Each copy is shifted by integer multiples of the sampling frequency $k\omega_s$ and scaled in amplitude by $\frac{1}{T_s}$.



4 The Nyquist-Shannon Sampling Theorem & Aliasing

4.1 Statement of the Theorem

The Nyquist-Shannon Sampling Theorem provides the exact condition under which a band-limited continuous-time signal can be perfectly reconstructed from its discrete samples without any loss of information.

Nyquist-Shannon Sampling Theorem

Let $x(t)$ be a continuous-time signal band-limited to ω_M, such that its Fourier transform satisfies $X(j\omega) = 0$ for $|\omega| > \omega_M$. The signal $x(t)$ can be uniquely and perfectly recovered from its sampled version $x_p(t) = \sum x(nT_s)\delta(t - nT_s)$ if and only if the sampling frequency $\omega_s = \frac{2\pi}{T_s}$ satisfies:

$$\omega_s > 2\omega_M \quad \Longleftrightarrow \quad f_s > 2f_M \tag{27}$$

The boundary value $2\omega_M$ is termed the **Nyquist rate**, while half the sampling frequency ($\omega_s/2$) is called the **Nyquist frequency** or *folding frequency*.

Real-World Example (Audio Processing): The human audible frequency spectrum spans approximately 20 Hz to 20 kHz ($f_M = 20$ kHz). According to the Nyquist theorem, audio signals must be sampled at a rate strictly greater than 2×20 kHz $= 40$ kHz. In digital audio engineering, the standard sampling frequency for CDs and uncompressed audio is set to 44.1 kHz (or 48 kHz), providing a safe guard band above 40 kHz to account for non-ideal anti-aliasing filter roll-off.

4.2 Geometric Intuition & Spectral Overlapping

Recall the expression for the spectrum of an impulse-sampled signal:

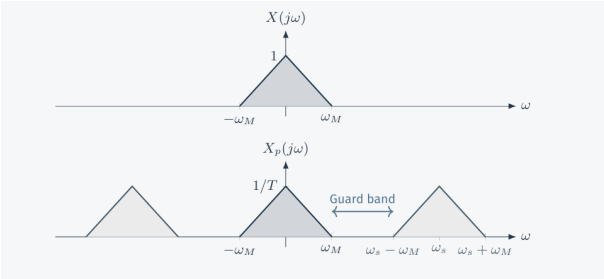
$$X_p(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)) \tag{28}$$

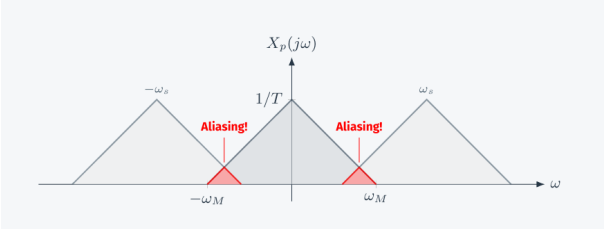
The spectrum $X_p(j\omega)$ consists of identical copies of $X(j\omega)$ centered at $0, \pm\omega_s, \pm 2\omega_s, \ldots$, scaled by $\frac{1}{T_s}$.

- The baseband spectrum copy ($k = 0$) occupies the frequency interval $[-\omega_M, \omega_M]$.
- The first positive shifted spectral copy ($k = 1$) occupies the frequency interval $[\omega_s - \omega_M, \omega_s + \omega_M]$.

To prevent distortion, adjacent spectral copies must not overlap. Thus, the lower boundary of the first shifted copy must be strictly greater than the upper boundary of the baseband spectrum:

$$\omega_s - \omega_M > \omega_M \implies \omega_s > 2\omega_M \tag{29}$$





4.3 The Aliasing Phenomenon

When the sampling condition is violated ($\omega_s \leq 2\omega_M$), the tail of the shifted spectrum $X(j(\omega - \omega_s))$ overlaps with the high-frequency portion of the baseband spectrum $X(j\omega)$. This overlapping phenomenon is called **aliasing**.

When aliasing occurs, high-frequency components of the original signal masquerade as lower-frequency components in the sampled spectrum. Summing these overlapping regions irreversibly corrupts the baseband spectrum. Consequently, no continuous-time filter can isolate $X(j\omega)$ from $X_p(j\omega)$, rendering perfect reconstruction impossible.

5 Signal Reconstruction

Assuming the Nyquist rate condition $\omega_s > 2\omega_M$ is satisfied, there is no spectral overlap. We now consider the problem of recovering the original continuous-time signal $x(t)$ from the sampled spectrum $X_p(j\omega)$.

5.1 Ideal Reconstruction in the Frequency Domain

Because $X_p(j\omega)$ contains periodically replicated copies of $X(j\omega)$ scaled by $\frac{1}{T_s}$, we can isolate the baseband spectrum by passing $X_p(j\omega)$ through an ideal continuous-time Continuous Low-Pass Filter (LPF) $H(j\omega)$.

To restore the amplitude scaling from $\frac{1}{T_s}$ back to 1, the filter must possess a passband gain of T_s. The ideal frequency response $H(j\omega)$ is defined as:

$$H(j\omega) = \begin{cases} T_s & \text{for } |\omega| \leq \omega_c \\ 0 & \text{for } |\omega| > \omega_c \end{cases} \tag{30}$$

where the cutoff frequency ω_c is selected within the transition guard band:

$$\omega_M < \omega_c < \omega_s - \omega_M \quad \left(\text{typically chosen as } \omega_c = \frac{\omega_s}{2}\right) \tag{31}$$

5.2 The Need for Time-Domain Filtering in Real-Time Systems

Frequency-domain filtering operates on the entire signal spectrum at once. However, real-time physical systems cannot wait for an entire signal duration to complete before starting processing.

Consider a live presentation in a seminar hall: if signal processing were performed strictly in the frequency domain, the system would need to capture the speaker's entire half-hour speech, compute its global Fourier transform, apply $H(j\omega)$, and run the Inverse Continuous-Time Fourier Transform (ICTFT) before emitting any sound. The audience would hear nothing during the speech and receive the processed output two hours later!

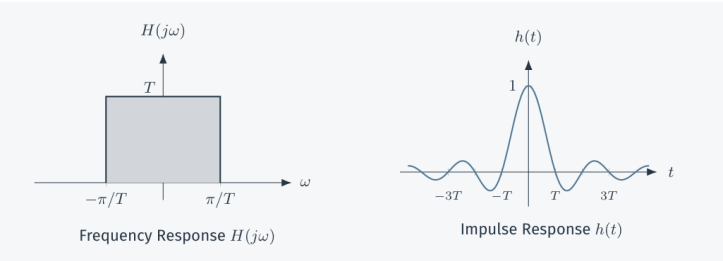
Real-time processing requires causal, continuous time-domain operations. By the convolution property, multiplication in the frequency domain translates to continuous-time convolution:

$$x(t) = x_p(t) * h(t) \tag{32}$$

5.3 Whittaker-Shannon Interpolation (Sinc Filter)

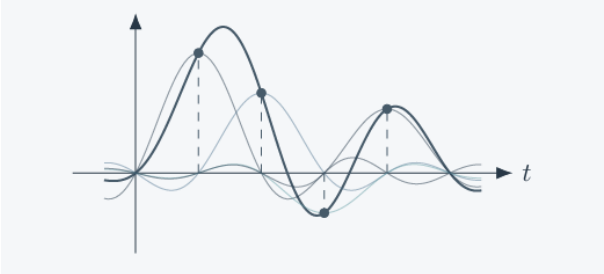
To derive the time-domain impulse response $h(t)$ of the ideal low-pass filter, we take the ICTFT of $H(j\omega)$ with cutoff $\omega_c = \frac{\omega_s}{2} = \frac{\pi}{T_s}$:

$$\begin{aligned} h(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\omega) e^{j\omega t}\, d\omega \\ &= \frac{1}{2\pi} \int_{-\pi/T_s}^{\pi/T_s} T_s e^{j\omega t}\, d\omega \\ &= \frac{T_s}{2\pi j t} \left[e^{j\frac{\pi}{T_s}t} - e^{-j\frac{\pi}{T_s}t} \right] \\ &= \frac{T_s}{\pi t} \sin\left(\frac{\pi t}{T_s}\right) = \operatorname{sinc}\left(\frac{t}{T_s}\right) \end{aligned} \tag{33}$$



Substituting $x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$ into the time-domain convolution yields the celebrated **Whittaker-Shannon Interpolation Formula**:

$$\begin{aligned} x(t) &= x_p(t) * h(t) \\ &= \left(\sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s) \right) * \operatorname{sinc}\left(\frac{t}{T_s}\right) \\ &= \sum_{n=-\infty}^{\infty} x(nT_s) \cdot \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right) \end{aligned} \tag{34}$$



5.4 Practical Limitations of the Ideal Filter

While Whittaker-Shannon interpolation guarantees mathematically perfect reconstruction, implementing an ideal sinc reconstruction filter in physical hardware is impossible due to fundamental physical constraints:

1. **Non-Causality & Infinite Delay:** The sinc function $\text{sinc}(t/T_s)$ extends infinitely into both the positive and negative time directions ($t \to -\infty$). Generating the output $x(t)$ at the present time requires knowledge of all future samples $x(nT_s)$ up to $t = +\infty$.

2. **Unattainability of Perfect Band-Limiting:** By the uncertainty principle of signal analysis (and the Paley-Wiener theorem), a signal cannot be strictly time-limited and strictly frequency-band-limited simultaneously. Because physical signals are time-limited (they have a beginning and an end), they can never be perfectly band-limited.

Since ideal frequency response brick-wall filters cannot be built in practice, practical systems must adopt causal continuous-time approximations, as explored in the following sections.

6 Practical Approximations to Ideal Reconstruction

Because ideal sinc interpolation requires a non-causal, infinitely long time-domain response, practical digital-to-analog converters (DACs) must approximate the ideal low-pass filter $H(j\omega)$ using finite, causal continuous-time operations.

To evaluate any practical reconstruction scheme:

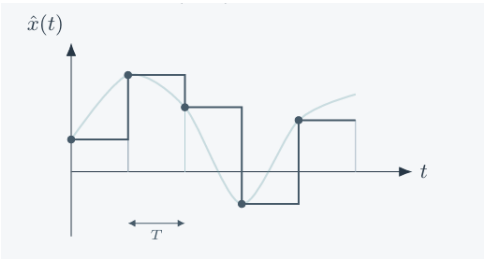
1. We define its time-domain impulse response $h(t)$ (how it interpolates discrete samples in time).

2. We compute its continuous-time Fourier transform $H(j\omega)$ (frequency response).

3. We compare $H(j\omega)$ against the ideal rectangular low-pass filter $H_{\text{ideal}}(j\omega)$ to evaluate passband distortion and high-frequency attenuation.

6.1 Zero-Order Hold (ZOH)

The **Zero-Order Hold (ZOH)** is the simplest and most widely used practical reconstruction filter. It operates by holding the value of each discrete sample $x(nT_s)$ constant until the subsequent sample arrives at $t = (n+1)T_s$.



6.1.1 Time-Domain Representation

In the time domain, the impulse response $h_{\text{ZOH}}(t)$ is a causal rectangular pulse of amplitude 1 and duration T_s:

$$h_{\text{ZOH}}(t) = \begin{cases} 1 & \text{for } 0 \leq t < T_s \\ 0 & \text{otherwise} \end{cases} \tag{35}$$

When an impulse-sampled signal $x_p(t) = \sum x(nT_s)\delta(t - nT_s)$ passes through $h_{\text{ZOH}}(t)$, convolution produces a piecewise-constant continuous-time staircase waveform $\tilde{x}(t)$:

$$\tilde{x}(t) = x_p(t) * h_{\text{ZOH}}(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\, p_{\text{rect}}(t - nT_s) \tag{36}$$

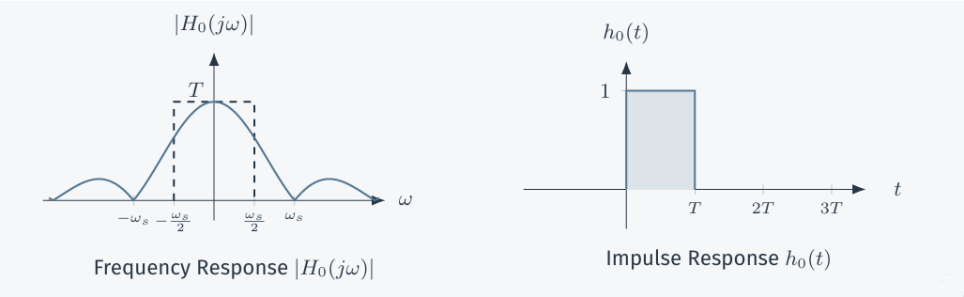
6.1.2 Frequency Response Analysis

Taking the continuous-time Fourier transform of $h_{\text{ZOH}}(t)$:

$$\begin{aligned} H_{\text{ZOH}}(j\omega) &= \int_0^{T_s} (1)e^{-j\omega t}\, dt \\ &= \frac{1}{-j\omega}\left[e^{-j\omega T_s} - 1\right] \\ &= e^{-j\omega T_s/2}\left[\frac{e^{j\omega T_s/2} - e^{-j\omega T_s/2}}{j\omega}\right] \\ &= T_s e^{-j\omega T_s/2} \cdot \frac{\sin(\omega T_s/2)}{\omega T_s/2} \end{aligned} \tag{37}$$

Expressed in terms of the normalized sinc function:

$$H_{\text{ZOH}}(j\omega) = T_s e^{-j\omega T_s/2}\, \text{sinc}\left(\frac{\omega T_s}{2\pi}\right) \tag{38}$$



6.1.3 Performance & Trade-offs

Comparing $|H_{\text{ZOH}}(j\omega)|$ with the ideal rectangular brick-wall filter reveals two significant real-world limitations:

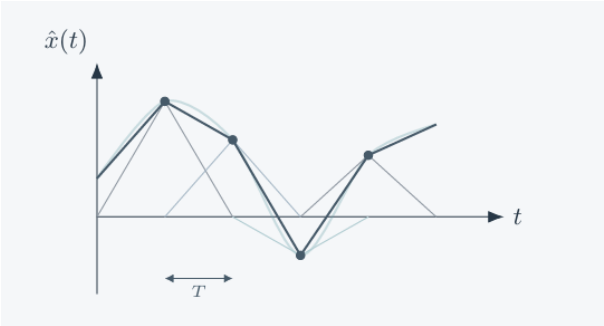
1. **High-Frequency Leakage (Imaging Artifacts):** Unlike the ideal LPF which strictly zeroes out all frequencies above $|\omega| > \omega_s/2$, the sinc function decays slowly ($1/\omega$). The side lobes of $H_{\text{ZOH}}(j\omega)$ extend into higher frequency regions, leaving residual high-frequency spectral replicas (imaging distortion) in the reconstructed signal.

2. **In-Band Passband Droop:** Within the desired passband $|\omega| \leq \omega_s/2$, $H_{\text{ZOH}}(j\omega)$ does not maintain a constant gain of T_s. Instead, it gradually rolls off, reaching a attenuation factor of $\frac{2}{\pi} \approx 0.637$ (-3.92 dB) at the Nyquist frequency $\omega = \omega_s/2$. This attenuates higher-frequency components within the baseband signal.

Despite these limitations, ZOH is universally implemented in electronic DACs because it requires no prior or future sample knowledge. In practice, a secondary continuous-time analog filter (called an *anti-imaging filter* or *equalizer*) is placed after the DAC to flatten the in-band gain and suppress side lobes.

6.2 First-Order Hold (FOH)

The **First-Order Hold (FOH)** improves upon ZOH by performing linear interpolation between consecutive sample points. Instead of holding a constant value, FOH connects adjacent samples with straight-line segments.



6.2.1 Time-Domain Representation

To form straight line segments centered at each sample instant, the non-causal symmetric impulse response $h_{\text{FOH}}(t)$ is defined as a triangular pulse of width $2T_s$ and peak height 1:

$$h_{\text{FOH}}(t) = \begin{cases} 1 - \frac{|t|}{T_s} & \text{for } |t| \leq T_s \\ 0 & \text{otherwise} \end{cases} \tag{39}$$

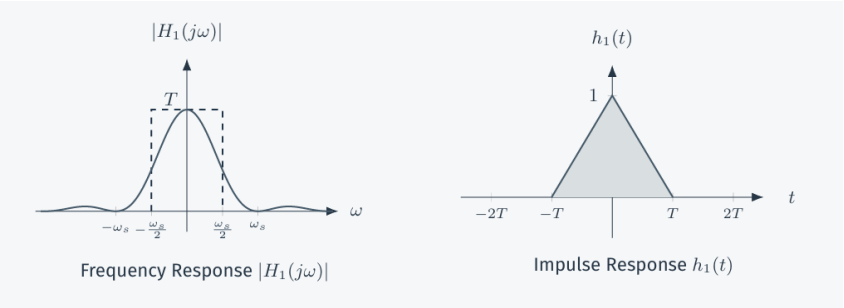
6.2.2 Frequency Response Analysis

Remarkably, a continuous triangular pulse $h_{\text{FOH}}(t)$ can be generated by convolving two centered rectangular ZOH pulses of width T_s:

$$h_{\text{FOH}}(t) = \frac{1}{T_s}\Big[h_{\text{ZOH, sym}}(t) * h_{\text{ZOH, sym}}(t)\Big] \tag{40}$$

By the convolution property of Fourier transforms, convolution in time corresponds to multiplication in frequency:

$$H_{\text{FOH}}(j\omega) = T_s \,\text{sinc}^2\left(\frac{\omega T_s}{2\pi}\right) \tag{41}$$



6.3 Comparison & Higher-Order Hold Cascades

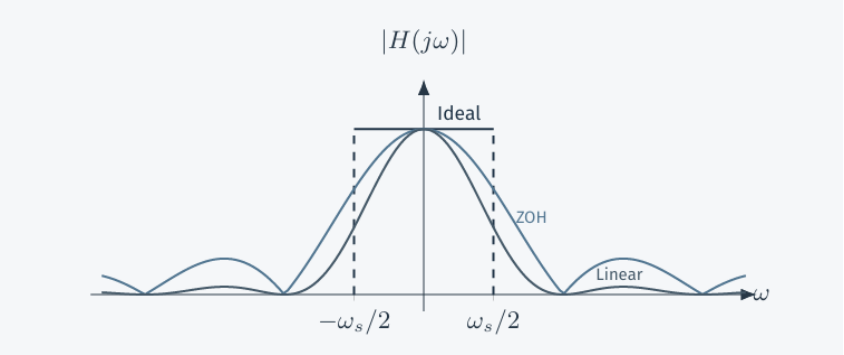
By repeatedly convolving n zero-order hold pulses, we can generate n-th order holds ($2^{\text{nd}}, 3^{\text{rd}}, \ldots, n^{\text{th}}$ order holds), whose frequency responses correspond to higher powers of the sinc function:

$$H^{(n)}(j\omega) \propto \text{sinc}^{n+1}\left(\frac{\omega T_s}{2\pi}\right) \tag{42}$$

Fundamental Trade-off in Higher-Order Approximations

As the order n of the hold filter increases:

- **Superior High-Frequency Attenuation:** The side lobes decay faster ($\frac{1}{\omega^{n+1}}$), rapidly suppressing unwanted high-frequency spectral replicas outside the baseband.
- **Increased Passband Droop:** The main lobe narrows faster within $|\omega| \leq \omega_s/2$, causing heavier attenuation of high baseband frequencies.
- **Increased Delay & Complexity:** Higher-order holds require storage of multiple past/future samples, introducing physical latency into real-time systems.



Engineering Decision Matrix: If the priority is to strictly remove high-frequency imaging artifacts (e.g., in high-fidelity audio or high-precision control systems), higher-order hold filters paired with post-equalization are preferred. If low computational latency and minimum passband amplitude distortion are required without adding complex post-filters, lower-order holds like ZOH remain the optimal choice.

7 Inevitability of Aliasing in Real-World Signals

In our theoretical formulation of the Nyquist-Shannon Sampling Theorem, we assumed that the continuous-time signal $x(t)$ is strictly band-limited—that is, its Fourier transform satisfies $X(j\omega) = 0$ for all $|\omega| > \omega_M$. However, this ideal assumption never holds true for physical, real-world signals.

7.1 The Time-Frequency Uncertainty Constraint

A fundamental principle of continuous-time signal analysis (governed by the Paley-Wiener theorem and the time-frequency uncertainty principle) dictates that **no non-zero signal can be simultaneously strictly time-limited and strictly frequency-band-limited**.

Because every real-world signal has a finite duration—it starts at a specific time and eventually ends—it is inherently time-limited. Consequently, its frequency spectrum $X(j\omega)$ extends infinitely across the frequency axis ($|\omega_M| \to \infty$).

This leads to a profound practical realization:

$$\text{Since } \omega_M = \infty, \quad \text{no finite sampling frequency } \omega_s \text{ can satisfy } \omega_s > 2\omega_M. \tag{43}$$

Therefore, **aliasing is mathematically unavoidable when sampling real-world physical signals**. Regardless of how high we set our sampling rate ω_s, high-frequency spectral tails will always fold back into the baseband spectrum.

7.2 Practical Workaround: Energy Containment & Effective Bandwidth

To bypass this limitation in practical engineering, we adopt an energy-based approximation:

1. **Defining Effective Bandwidth (ω_M):** Instead of requiring absolute band-limiting, we identify an *effective bandwidth* ω_M that captures the vast majority of the signal's total energy (typically 90%, 95%, or 99%). Energy outside this boundary is treated as negligible noise.

2. **Anti-Aliasing Filtering:** Before sampling, $x(t)$ is passed through an analog Continuous-Time Anti-Aliasing Low-Pass Filter (CT-AAF) with cutoff frequency $\omega_c \approx \omega_M$. This explicitly attenuates spectral components above ω_M before the sampling process takes place, keeping aliasing distortion below an acceptable threshold.

3. **Sampling at the Practical Nyquist Rate:** We then choose a sampling frequency $\omega_s > 2\omega_M$ based on this effective cutoff.

Practical Implications of Sampling Continuous Signals

Sampling a real-world physical signal fundamentally alters it forever. Because information beyond the effective bandwidth is filtered or folded back via aliasing, **perfect, exact mathematical reconstruction of a physical continuous signal is impossible**.
However, by carefully selecting an effective bandwidth ω_M and an appropriate sampling rate ω_s, the reconstructed continuous signal can be made imperceptibly close to the original. For human perception—whether viewing a high-refresh display or listening to high-fidelity audio—the reconstructed output is functionally indistinguishable from the original continuous-time signal.

8 Connection of Sampling with the DTFT and DFT

A core theme in signals and systems is the fundamental duality of the Fourier transform: **sampling in one domain induces periodicity in the other domain**.

- **Sampling in time $\longleftrightarrow$ Periodicity in frequency** (Continuous-Time Impulse Sampling)
- **Sampling in frequency $\longleftrightarrow$ Periodicity in time** (Discrete Fourier Transform)

This duality serves as the bridge between the Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT).

8.1 Mathematical Formulation

Consider a discrete-time sequence $x[n]$. Its Continuous-Frequency Discrete-Time Fourier Transform (DTFT) is defined as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \tag{44}$$

The spectrum $X(e^{j\omega})$ is continuous in ω and inherently periodic with period 2π.

The N-point Discrete Fourier Transform (DFT) evaluates a sequence of length N at discrete frequency points:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi kn}{N}}, \quad k = 0, 1, \ldots, N-1 \tag{45}$$

By defining N uniformly spaced frequencies along the continuous frequency axis over one 2π period:

$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \ldots, N-1 \tag{46}$$

we obtain the fundamental connection between the DTFT and the DFT:

DFT as a Frequency-Sampled DTFT

For a finite-duration sequence $x[n]$ supported on $n \in \{0, 1, \ldots, N-1\}$, the N-point DFT coefficients $X[k]$ are exact point samples of its continuous spectrum $X(e^{j\omega})$:

$$X[k] = X(e^{j\omega})\Big|_{\omega=\frac{2\pi k}{N}} \tag{47}$$

8.2 Frequency Sampling and Time-Domain Periodicity

Just as sampling a continuous signal in the time domain creates periodically repeated spectral copies in frequency ($X_p(j\omega) = \frac{1}{T_s}\sum X(j(\omega - k\omega_s))$), **sampling the continuous DTFT spectrum in frequency forces the time-domain sequence to become periodic**.

When we compute an N-point DFT, we implicitly construct a periodic extension $x_p[n]$ of the original finite sequence $x[n]$:

$$x_p[n] = x[n \pmod N] = \sum_{m=-\infty}^{\infty} x[n - mN] \tag{48}$$

For example, a 4-point sequence $\{1, 2, 3, 4\}$ processed by a 4-point DFT is implicitly extended to an infinite periodic sequence:

$$\ldots, 1, 2, 3, 4 \;\Big|\; 1, 2, 3, 4 \;\Big|\; 1, 2, 3, 4, \ldots \tag{49}$$

This implicit periodicity is made obvious by the Inverse Discrete Fourier Transform (IDFT):

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi kn}{N}} \tag{50}$$

Although the IDFT is computed from N discrete frequency points $X[k]$, evaluating the formula for any index n outside $\{0, \ldots, N-1\}$ yields $x[n+N] = x[n]$. Frequency discretization forces time-domain periodicity.

8.3 Summary of Fourier Duality Relationships

Transform	**Time Domain**	**Frequency Domain**
CTFT	Continuous, Aperiodic	Continuous, Aperiodic
CTFS	Continuous, Periodic	Discrete, Aperiodic
DTFT	Discrete, Aperiodic	Continuous, Periodic (2π)
DFT	Discrete, Periodic (N)	Discrete, Periodic (N)

Table 1: Duality of Continuity and Periodicity in Fourier Analysis

Relationship Between DTFT and DFT

The N-point DFT consists of N equally spaced samples of the DTFT at frequencies $\omega_k = \frac{2\pi k}{N}$. By Fourier duality, sampling the DTFT in frequency corresponds to making the time-domain sequence periodic with period N. Thus, the DFT is a frequency-sampled version of the DTFT of a periodic extension of the finite-length sequence.

All core DSP phenomena stem from two foundational duality principles:

$$\text{Truncation (Windowing)} \longleftrightarrow \text{Convolution} \quad \text{and} \quad \text{Sampling} \longleftrightarrow \text{Periodicity} \tag{51}$$

Understanding these dualities provides a unified framework for analyzing aliasing, spectral leakage, windowing, and digital filtering.

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